CatchPoint

Let them step. Catch only when they can’t.



ME5412 Robotics for Healthcare · Team project · 3 Nov 2026

Concept render (design D3, unbuilt): Blender; mannequin Mixamo/Adobe via three.js examples; HDRI Poly Haven (CC0).

Recovery steps prevent falls — but practising them is risky



71%

of stroke patients still had impaired reactive stepping at discharge

Inness et al., Phys Ther 2014

14.7%

fell during inpatient stroke rehabilitation

Mansfield et al., Neurorehabil Neural Repair 2013

684,000

deaths from falls every year worldwide (all ages, all causes)

World Health Organization

Photo: US Navy, public domain (Wikimedia Commons). Today therapists guard patients by hand, so they tend to over-support.

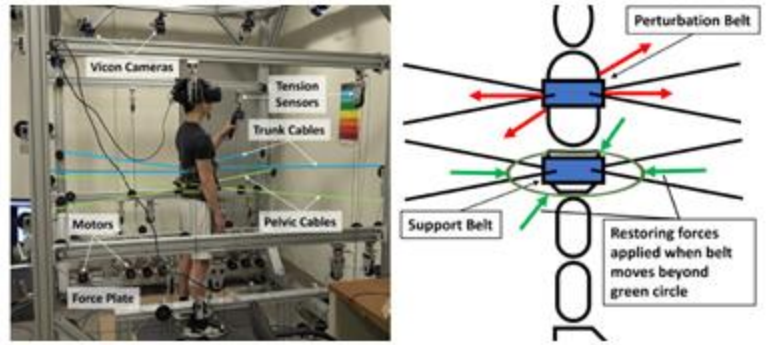
Threshold-triggered support acts too early or too late



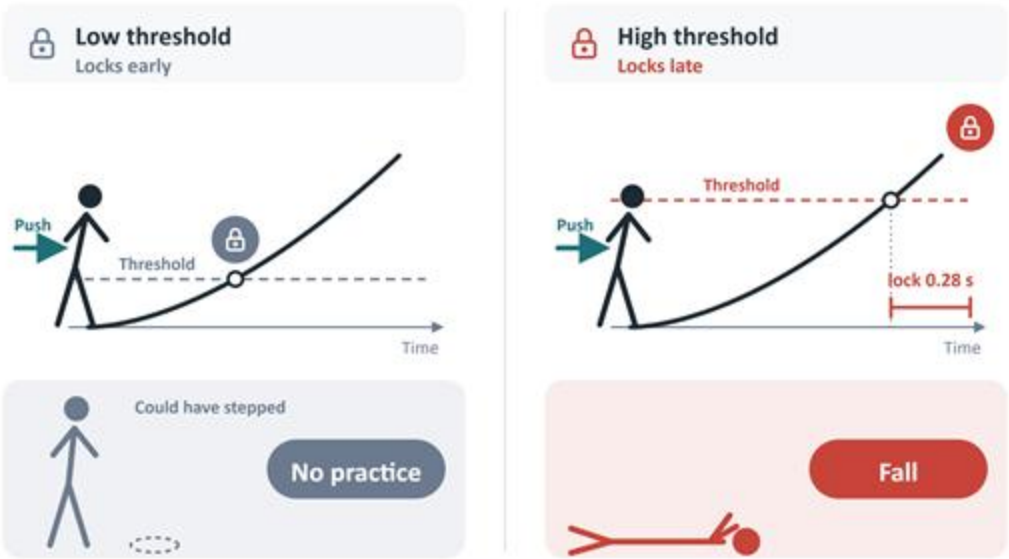
MRBA (NTU): brakes lock when a sensor threshold trips

Photo omitted in public version
(Hocoma Andago — see source link)

Andago (Hocoma): overground support walker



RobUST: pelvic force field outside a fixed boundary



Locks early — the patient never gets to practise stepping.

Locks late — the threshold trips late, and a 0.28 s lock may come too late.

Photos: NTU MRBA, J NeuroEng Rehabil 2023 (CC BY 4.0) · Hocoma Andago, © Hocoma/DIH · RobUST, Omofuma et al., Bioengineering 2023 (CC BY). Lock response: MRBA 0.28 s (ref. [1]).

What is new: CatchPoint vs existing systems

	Mobile overground	Predictive trigger XCoM + delay compensation	Patient-specific step reach	Graded minimum force	Preserves self-recovery practice	Catches only when needed	Nudge + catch in one cable path	Passive fail-safe force limit
NTU MRBA	●	○	○	○	○	○	○	?
NTU DRBA	?	○	○	○	○	○	○	?
Hocoma Andago	?	○	?	○	?	○	?	?
KineAssist	●	○	○	○	◑	○	○	?
RobUST (Columbia)	○	○	◑	?	●	○	◑	?
mTPAD (Columbia)	●	○	○	◑	?	○	○	?
ZeroG	◑	○	○	○	?	?	○	?
NIMBLE (URJC)	●	?	?	?	?	?	○	?
ExoRecovery (exo)	?	●	?	○	○	?	○	?
CatchPoint (ours)	◑	●	●	●	●	●	◑	◑

● yes ◑ partial ○ no ? not reported in sources we read

Based on published descriptions we could access; ? = not reported. CatchPoint: simulation + design only.

To our knowledge, none of the systems we read combines a predictive, per-patient trigger with a graded minimum force and a catch only when needed. Cells marked ? are not reported in the sources we read; we do not count them against a system.

Source: team comparison of published system descriptions we could access (per-cell sources in the notes). CatchPoint: simulation and design only.

Three contributions, each against its closest prior work

New control approach

Recoverability-based trigger

Act when the predicted capture point leaves this patient's own step reach.

Closest prior work
vs MRBA / DRBA / RobUST: fixed threshold or static boundary

Evidence: simulation; physics demo (re-calibrated region)

New control algorithm

Graded minimum nudge

Smallest pelvic force (≤ 150 N) that restores a reachable step; catch only if not enough.

Closest prior work
vs threshold-lock walkers (on/off) and always-on support

Evidence: simulation; physics demo (re-calibrated region)

New interface / actuation (design)

One cable set, three modes

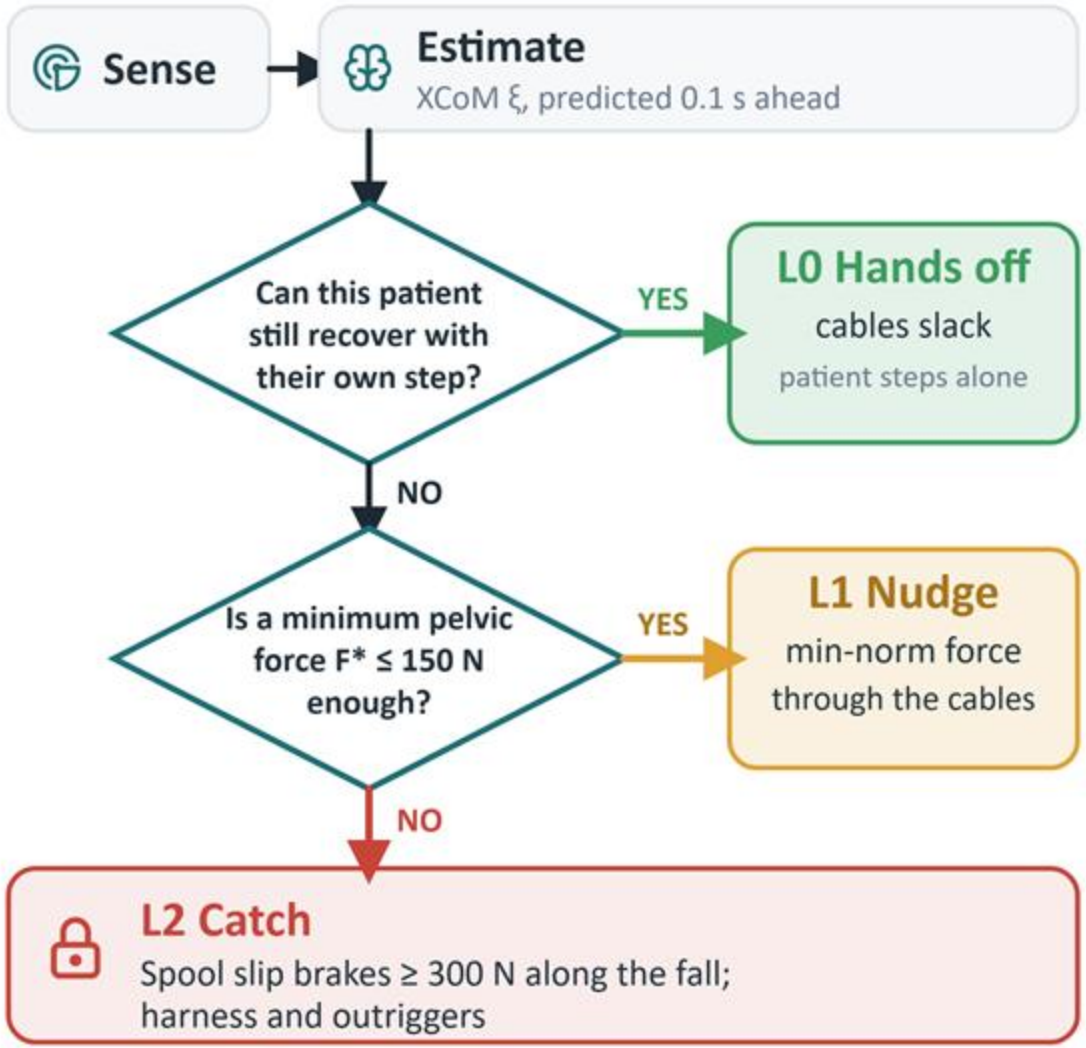
Servo nudges and power-off slip brakes stop the fall, on the same eight cables.

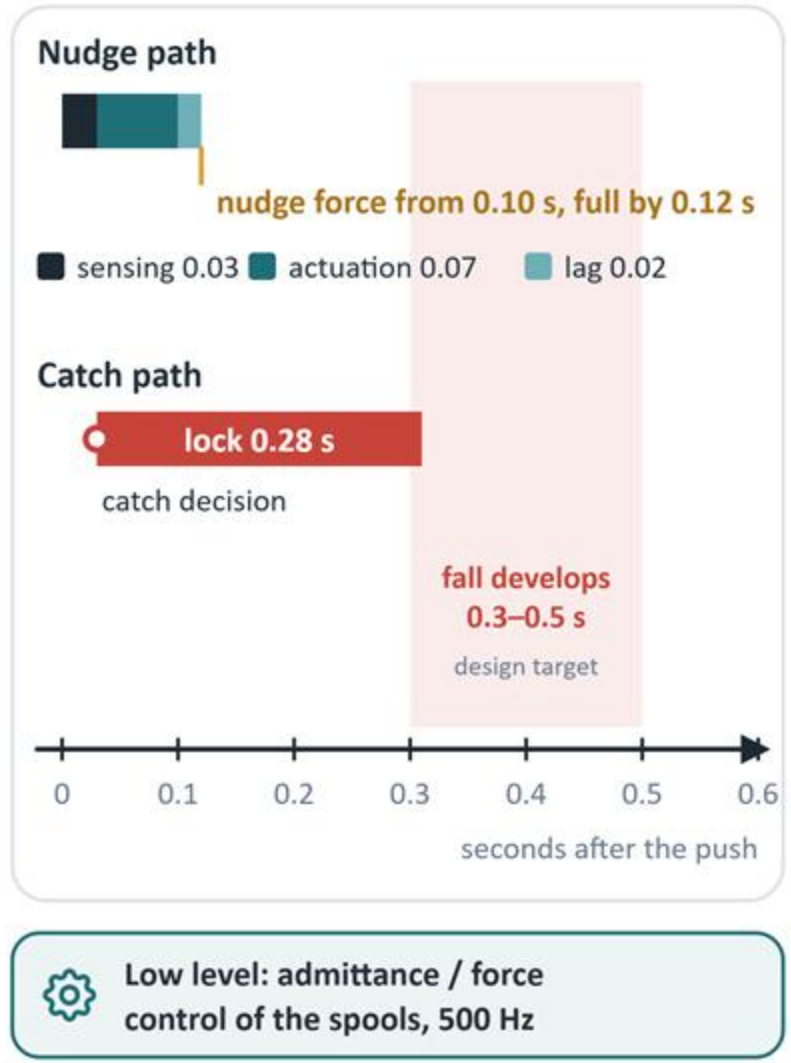
Closest prior work
vs mTPAD (nudge, no catch); overhead systems such as ZeroG (vertical catch only)

Evidence: design analysis; unbuilt

Categories follow the course briefing. Physics demo: calibrated on the same push grid; robot-policy demonstration only. Novelty "to our knowledge" (search-based, not proof).

Control architecture and timing





Two questions, in order

Recover alone? If not, is ≤ 150 N enough?

After a catch lock

COM stops 0.10 s later (median).

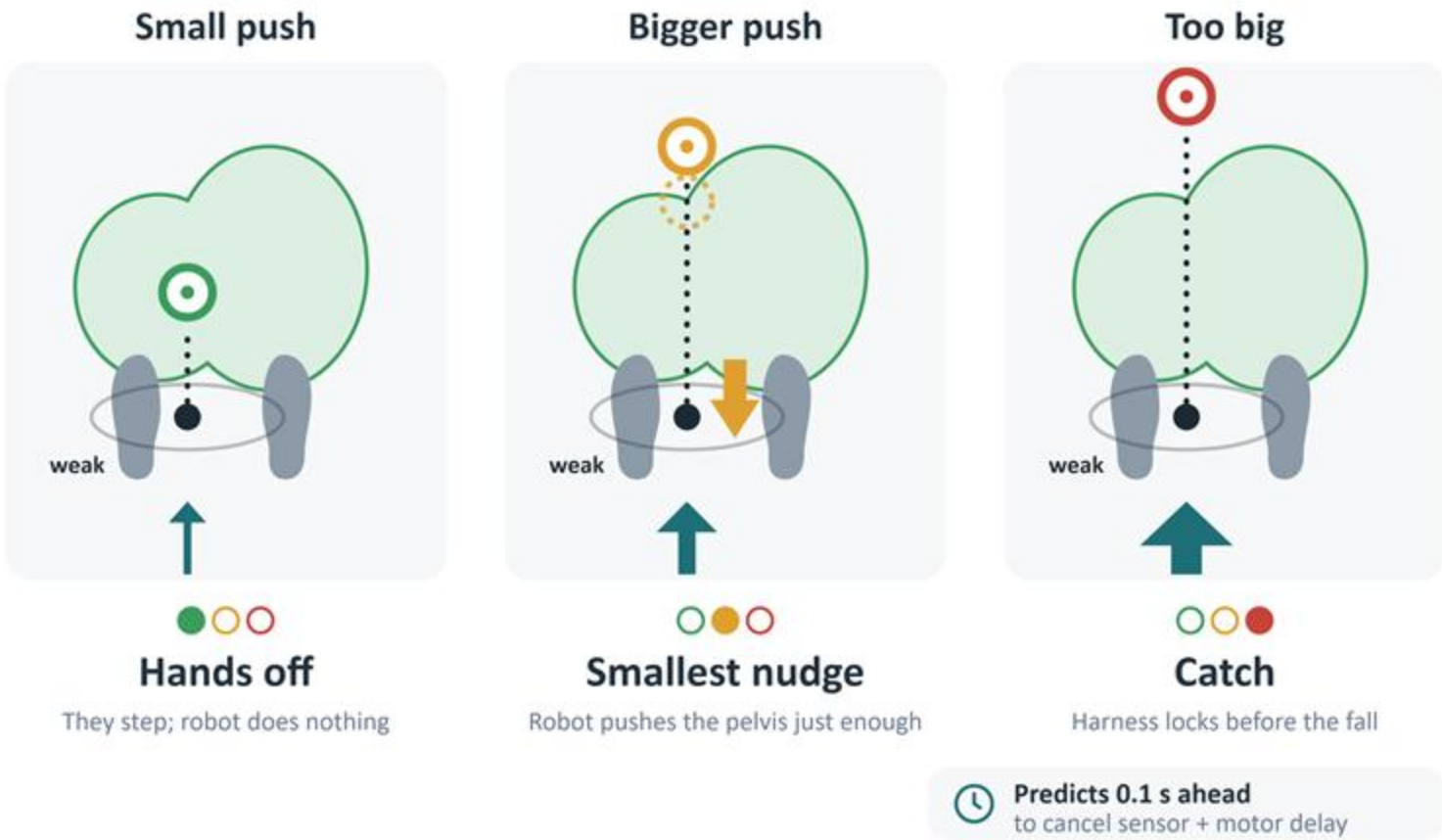
All delays

are model assumptions.

Figure by the project team. Delays (sensing 0.03 s, actuation 0.07 s, lag 0.02 s) are model assumptions, not measurements; COM stop times from the team's sizing simulation (passive slip brakes, lock 0.28 s).

Where they must step vs. where they can reach

Ask "can they still save themselves?" — not "have they started to fall?"



How to read it

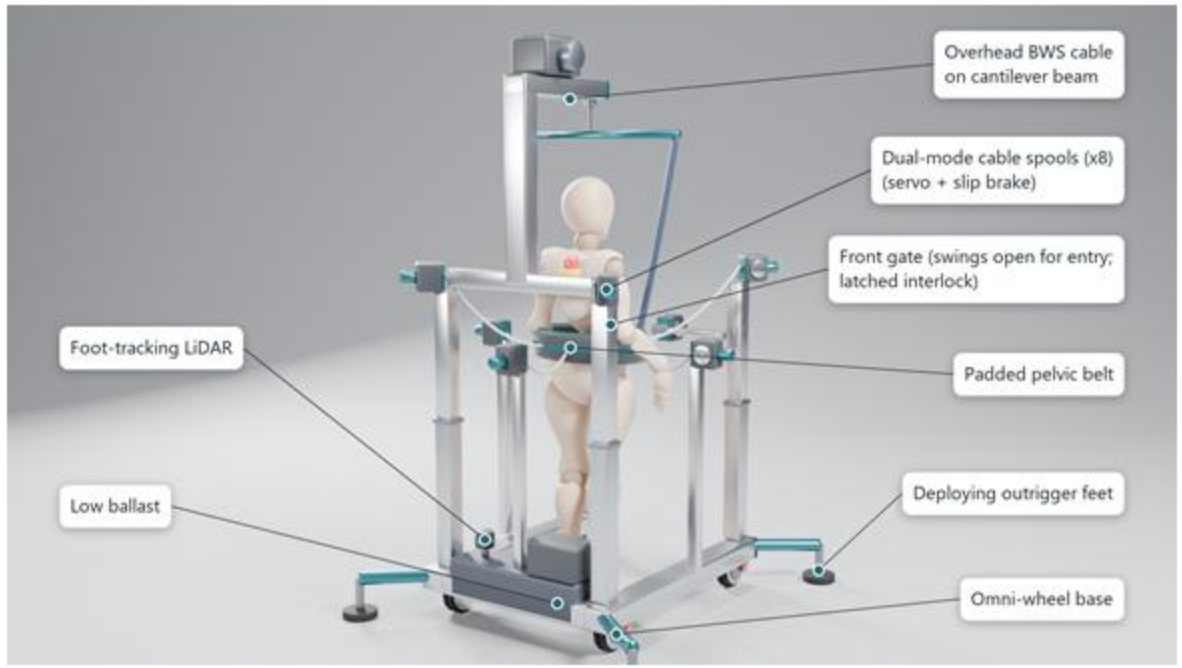
◎ = where they must step to stay upright (capture point)

Green area = where this patient's own step can reach

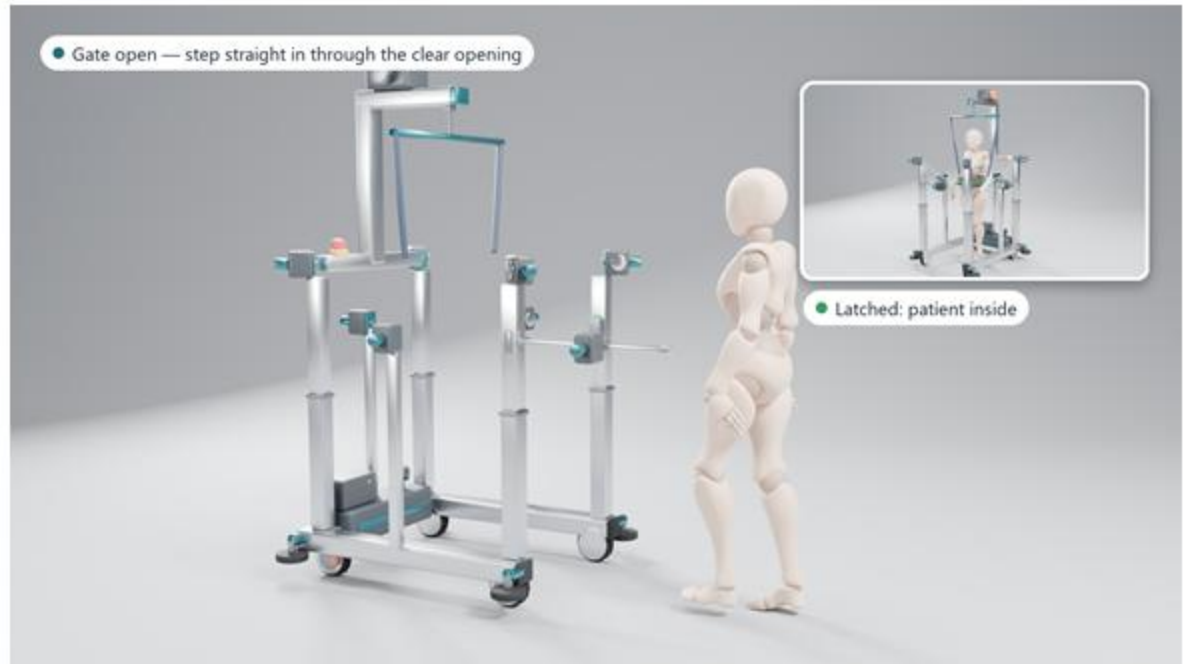
Inside it: hands off. Outside: smallest nudge. Out of reach: catch.

Schematic illustration, top view, by the project team. The capture point is a standard balance concept (Pratt 2006; Hof 2005).

Hardware: one cable set, three modes



Design D3: cables, dual-mode spools, overhead body-weight cable, outriggers



Latched front gate: opens to step in, latched during use

One cable set

Slack · nudge ≤ 150 N · catch ≥ 300 N

Dual-mode spools

Servo nudges; power-off slip brake catches

Designed not to tip

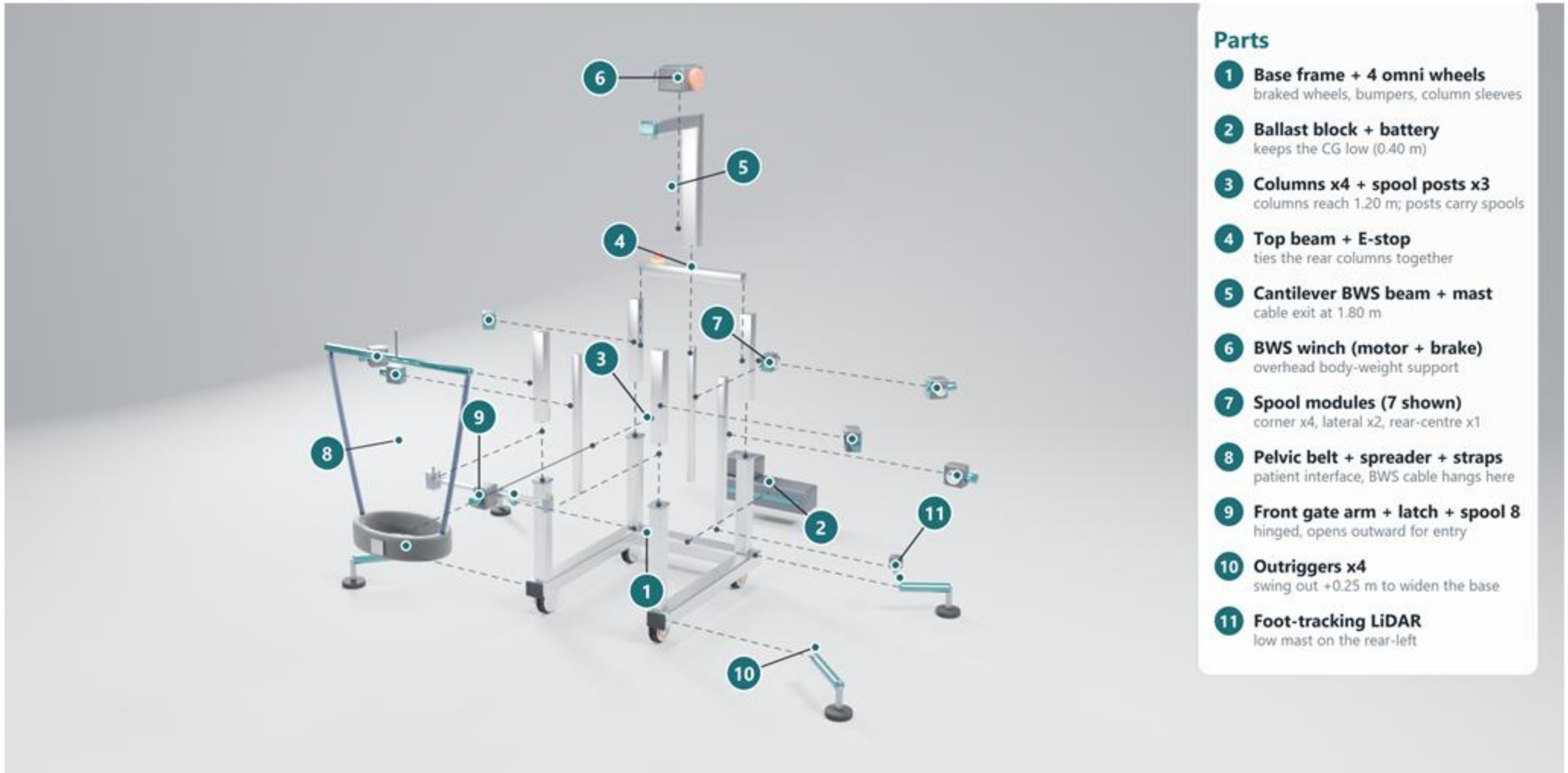
Tip SF ≥ 1.5 up to 420 N (statics)

Concept, unbuilt

Statics and simulation only

Render: Blender (design D3 CAD). Mannequin: Mixamo/Adobe via three.js examples; HDRI/texture: Poly Haven (CC0).

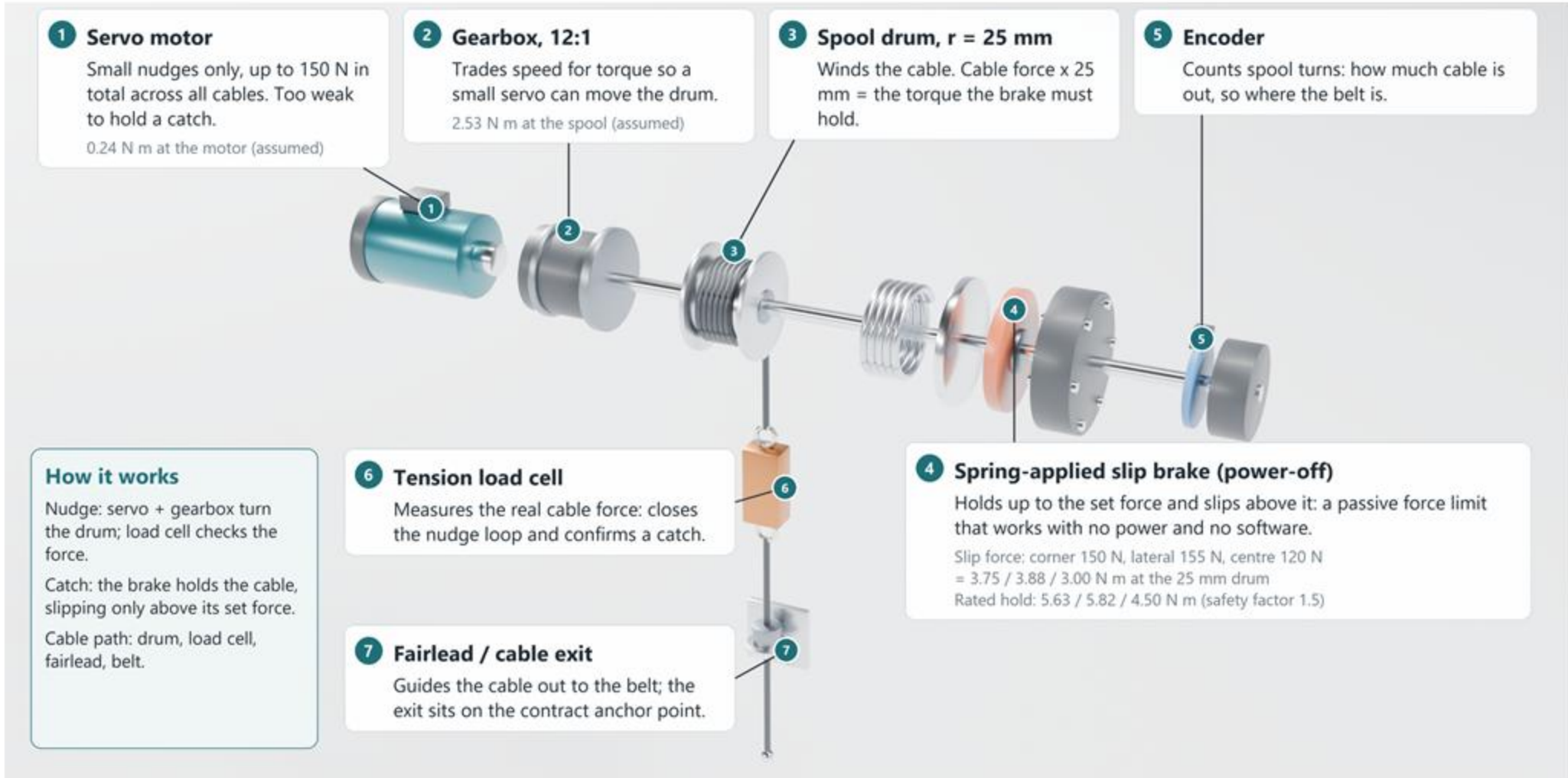
Design D3: exploded assembly



Parts pulled apart along their mounting axes; dashed lines show where each part mounts. Cables omitted for clarity. Concept render: unbuilt and unvalidated.

Render: Blender (design D3 CAD).

One dual-mode spool: servo for nudges, slip brake for the catch

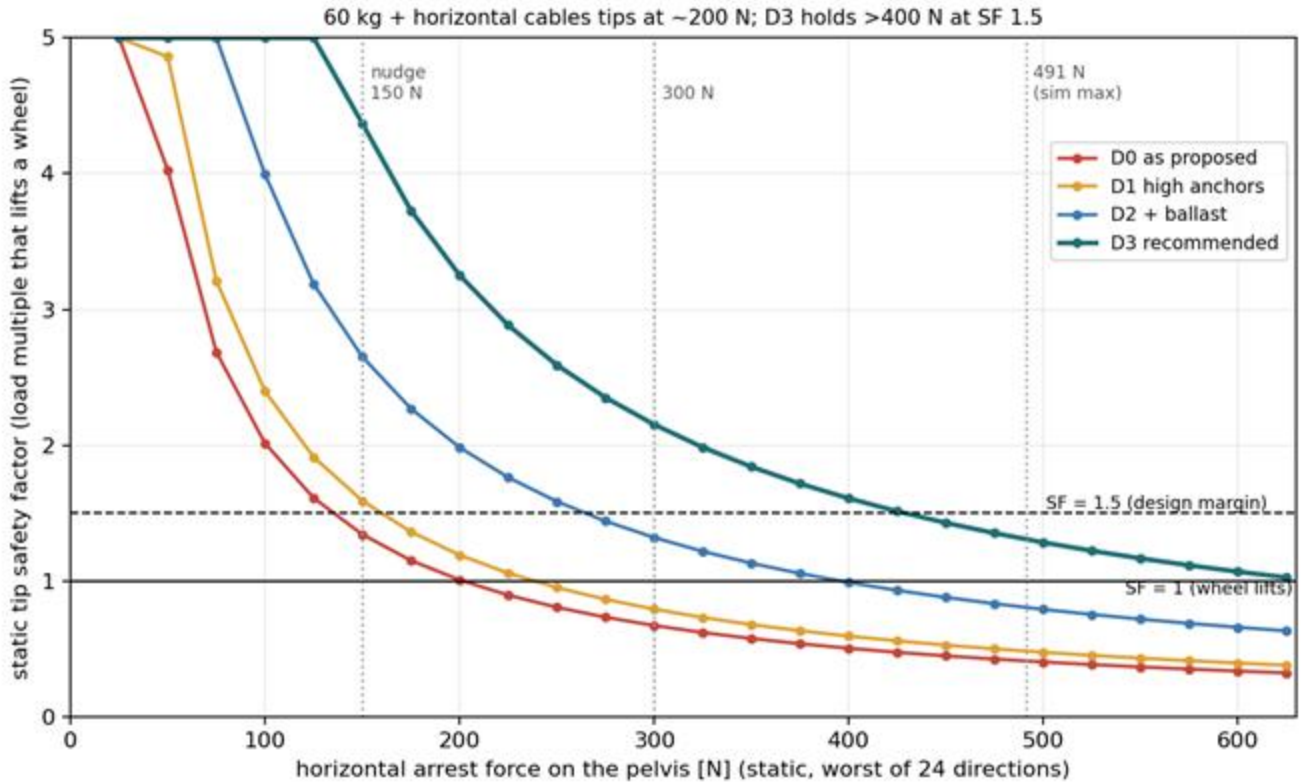


8 identical modules per walker; part shapes are placeholders, not a part selection. Concept render: unbuilt and unvalidated.

Render: Blender (design D3 CAD).

Can a walker catch without tipping?

Statics + simulation: an 80 kg low-CG walker with outriggers is not predicted to tip.



805/814

caught trials held by the passive slip brakes (98.8%, CI 97.9–99.5%)

377/814

same trials, original 60 kg design (46.3%)

380 ≤ 420 N

max slip-brake force vs tip limit (SF 1.5)

Not validated. Covers caught trials only; the other 1186/2000 were not caught (1091 recovered, 95 fell).

Source: team statics analysis of design D3 and one seeded 2,000-trial simulation (CatchPoint, lock 0.28 s). Static tip safety factor, worst of 24 directions.

Same push, three responses



Hands off: cables slack

Nudge: ≤ 150 N

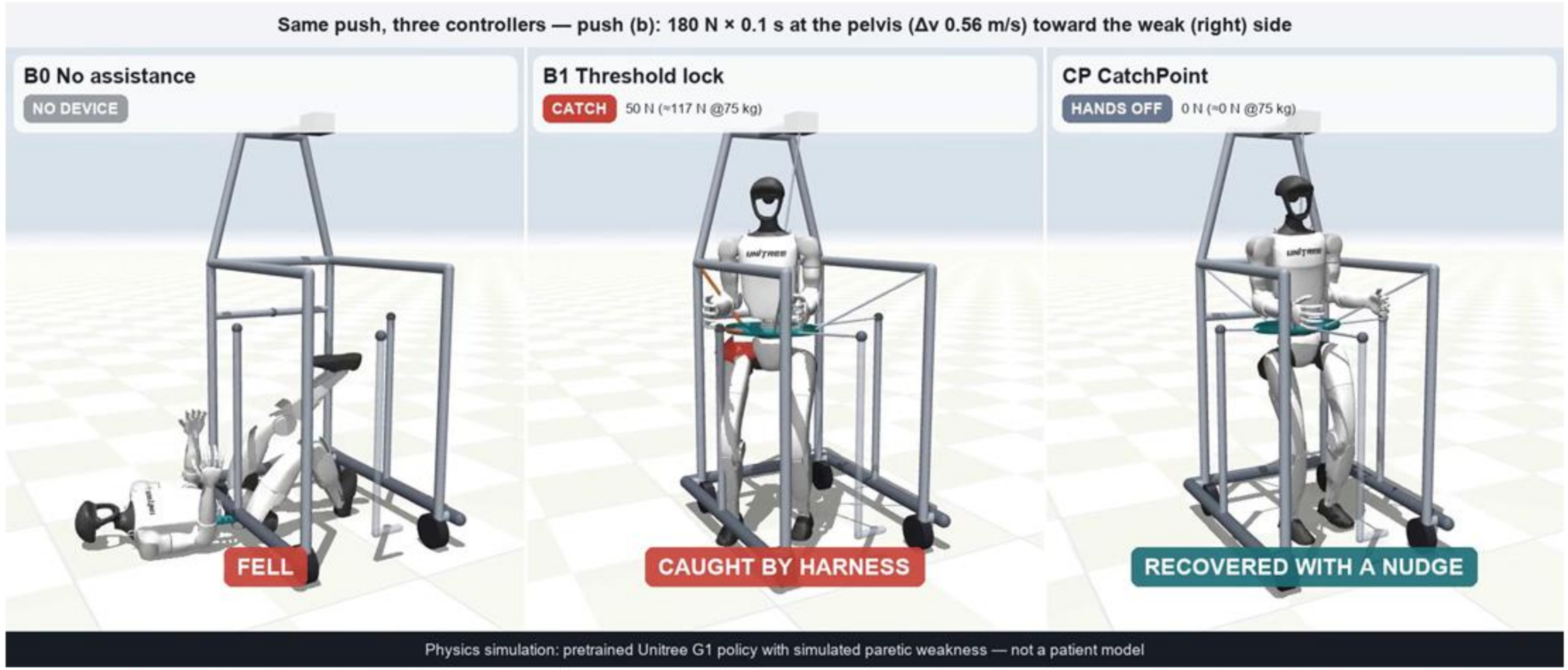
Catch: slip brakes ≥ 300 N along the fall, ≤ 380 N total

One cable set, three modes: slack · graded nudge ≤ 150 N · passive slip brakes: ≥ 300 N along the fall, ≤ 380 N total

Design limits. Concept render of design D3: unbuilt and unvalidated.

Render: Blender; mannequin Mixamo/Adobe via three.js examples; HDRI/texture Poly Haven (CC0).

The same push in physics simulation: fall, lock, or nudge

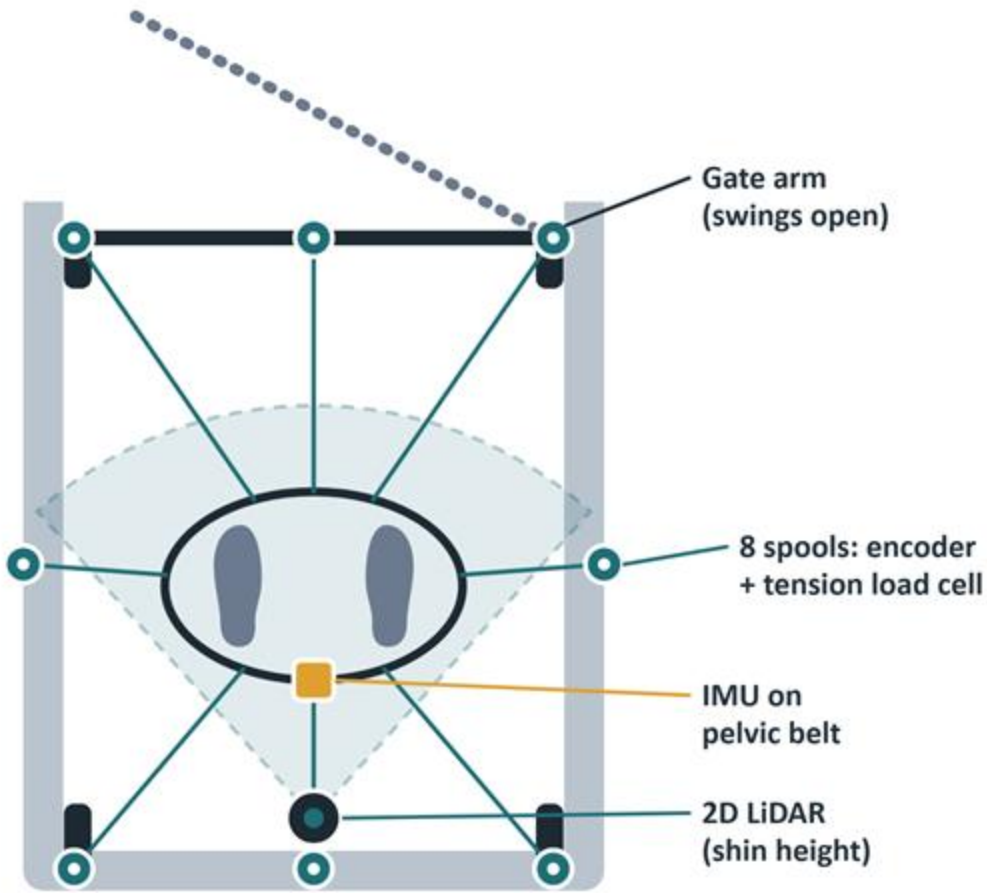


180 N push toward the weak side. No device: falls · threshold lock: harness catch · CatchPoint: a nudge (≤ 150 N).

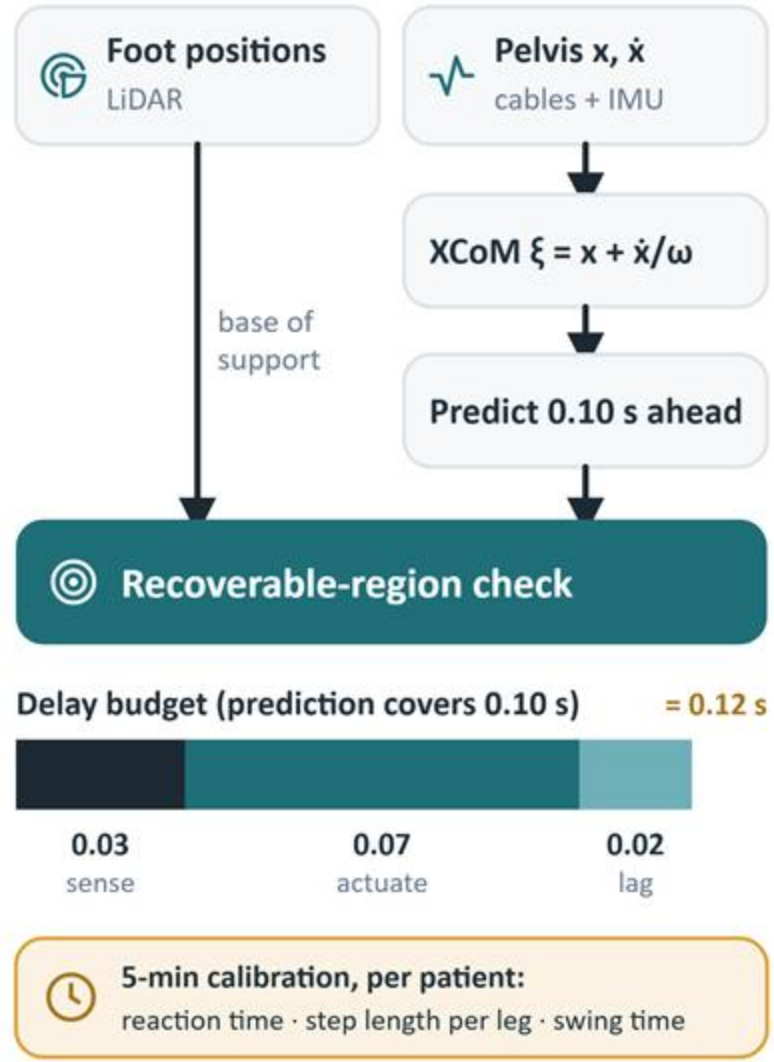
Robot-policy demonstration (Unitree G1 in MuJoCo), not a patient. Click the picture to play.

Source: MuJoCo physics demo, Unitree G1 policy with simulated weak right leg (one deterministic run).

Sensing and state estimation



D3 walker, top view · 0.95 × 0.80 m



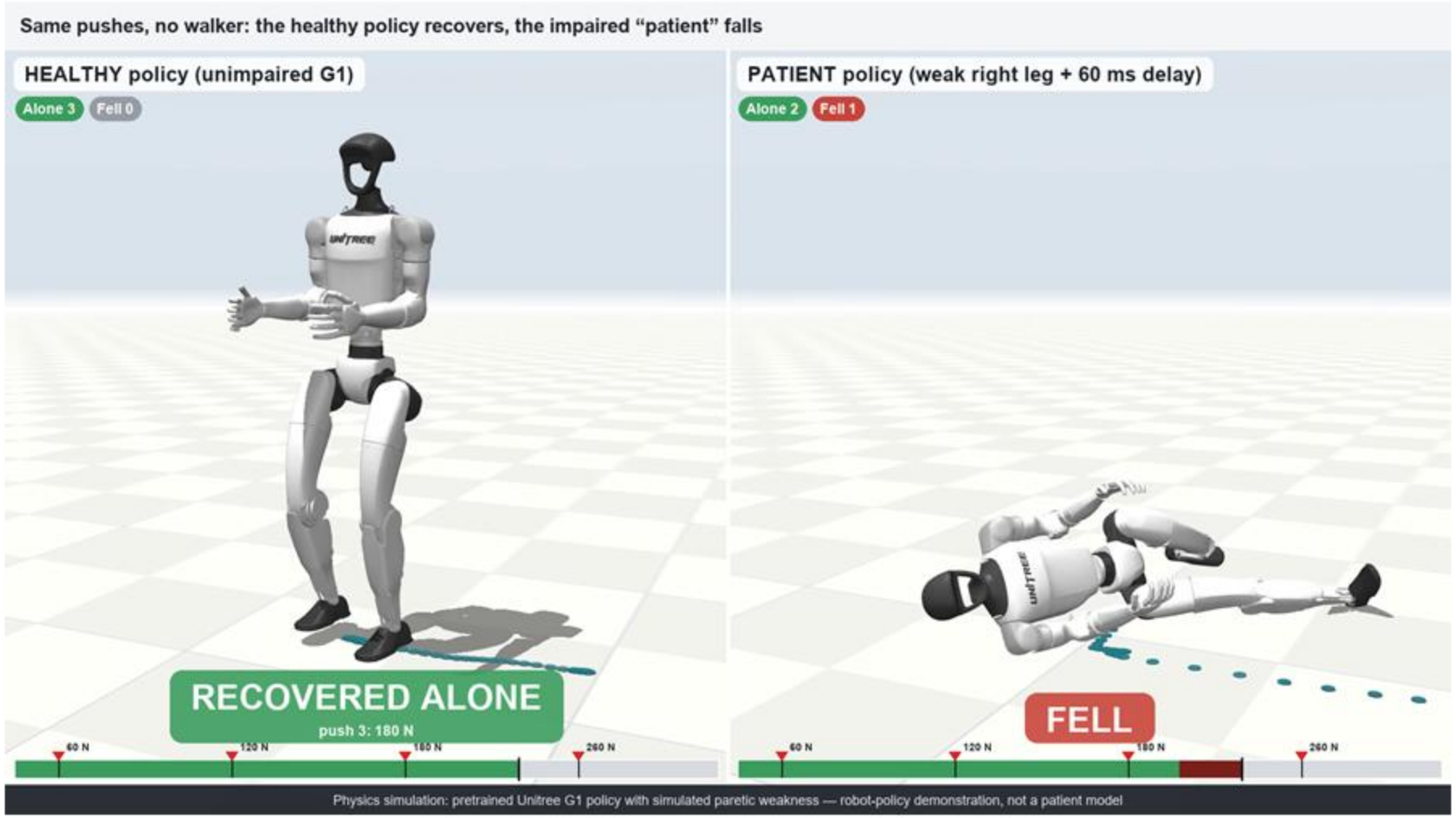
Sensed: feet, pelvis, cable force

Estimated: capture point (XCoM), 0.10 s ahead

Calibrated: 5 min per patient

Figure by the project team. Sensor choices are proposals, not tested hardware.

The physics patient: same pushes, healthy recovers, impaired falls



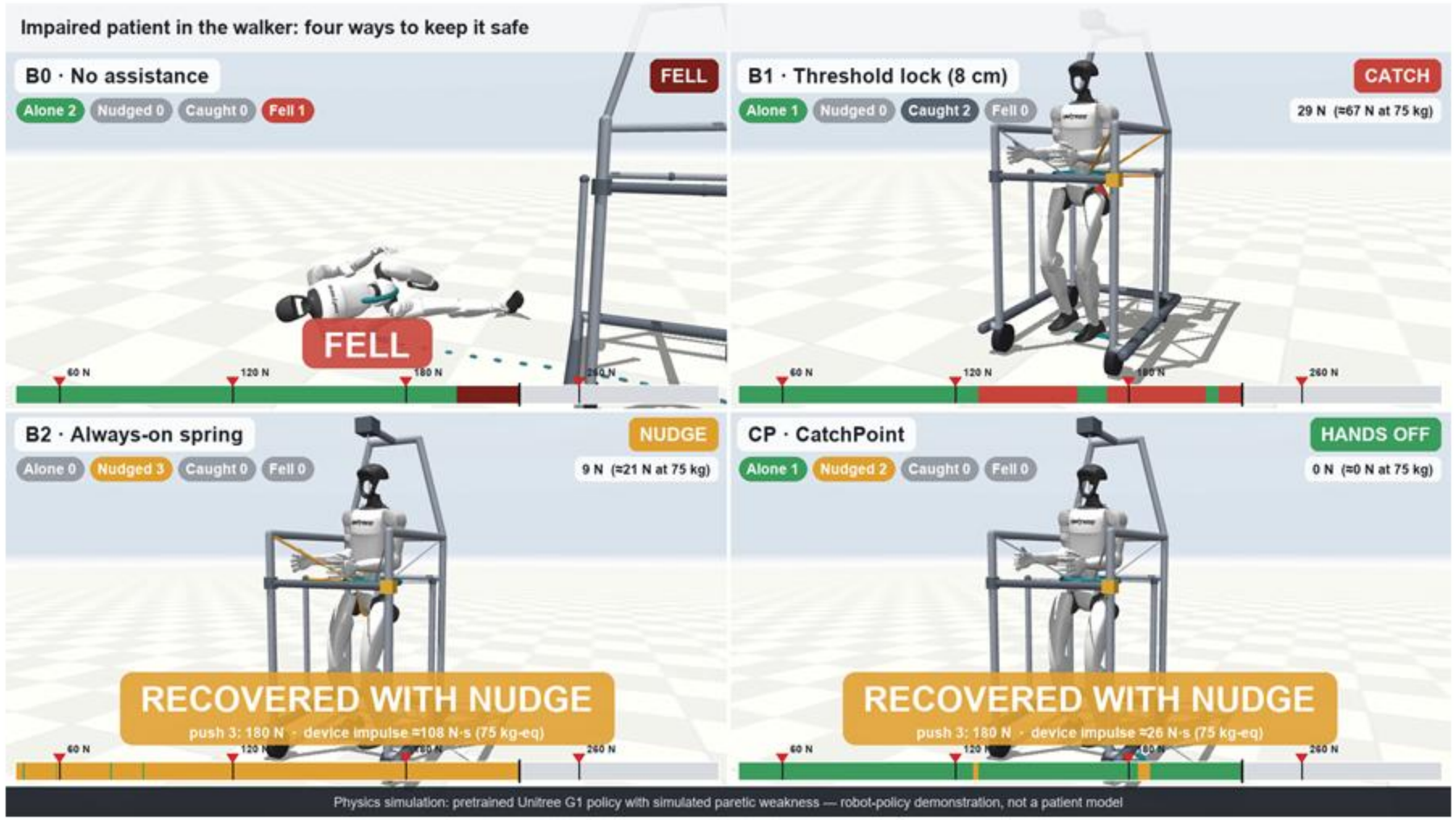
Pretrained Unitree G1 walking policy in MuJoCo.

Patient: right-leg torque ×0.5 plus 60 ms delay.

Four weak-side pushes (60 / 120 / 180 / 260 N): the healthy policy recovers from all four; the patient falls at 180 N.

Source: MuJoCo physics demo, Unitree G1 policy (one deterministic run). Robot-policy demonstration only — not a patient model. Click the picture to play.

Physics demo: fewer catch-locks, more nudges (≤ 150 N)

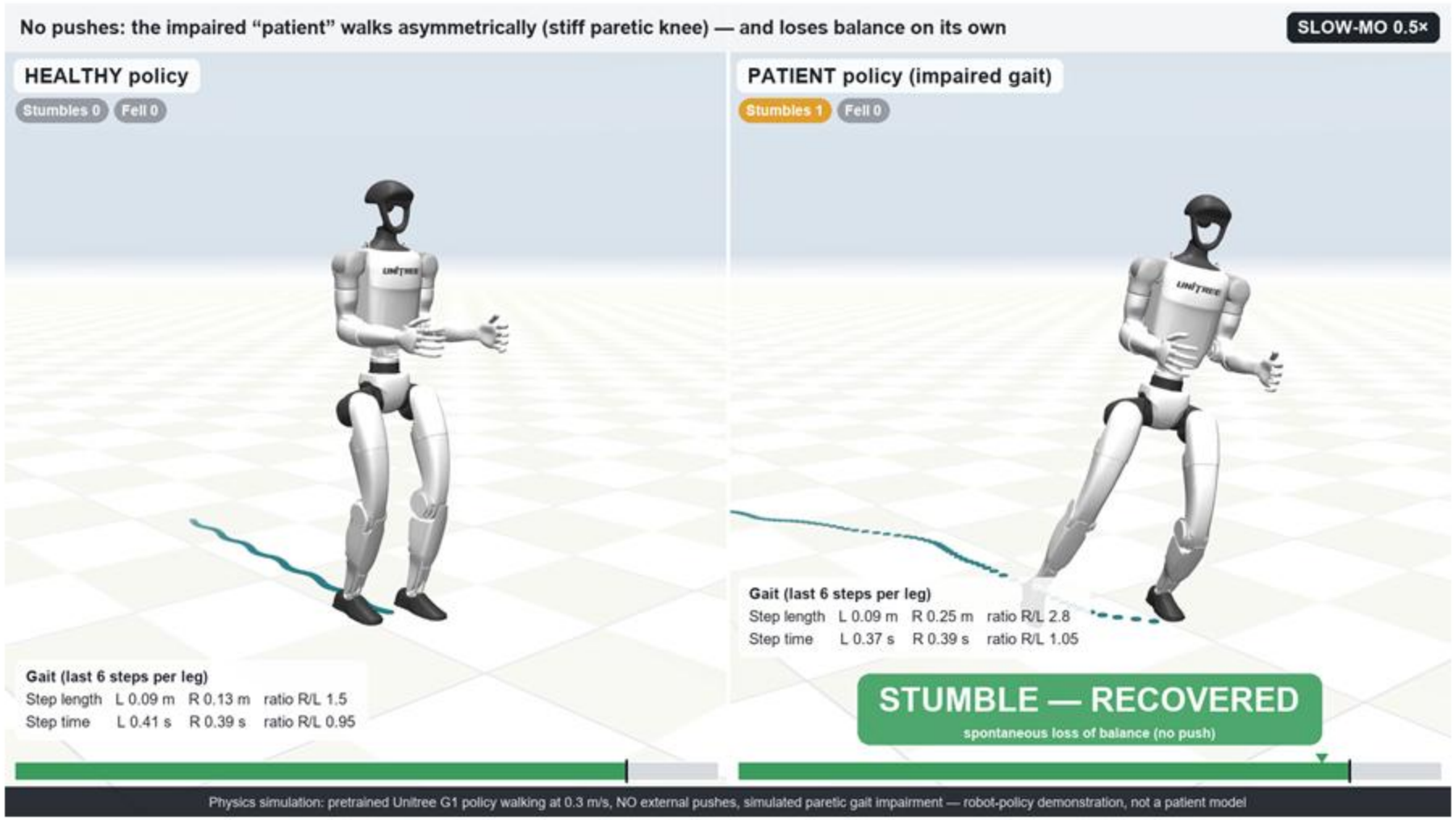


Same weak-side pushes, four ways: no robot, threshold lock (8 cm), always-on spring, CatchPoint.

Over the 56-push grid CatchPoint: 0 falls, 26 catch-locks, 23 nudges.

Source: MuJoCo physics demo, Unitree G1 policy (BSD-3). Robot-policy demonstration only — not patient evidence. Click the picture to play.

No push needed: the impaired gait shows on its own



Robot-policy demonstration only — not patient evidence. Click the picture to play.

Source: MuJoCo physics demo, Unitree G1 policy (walking, v3 protocol).

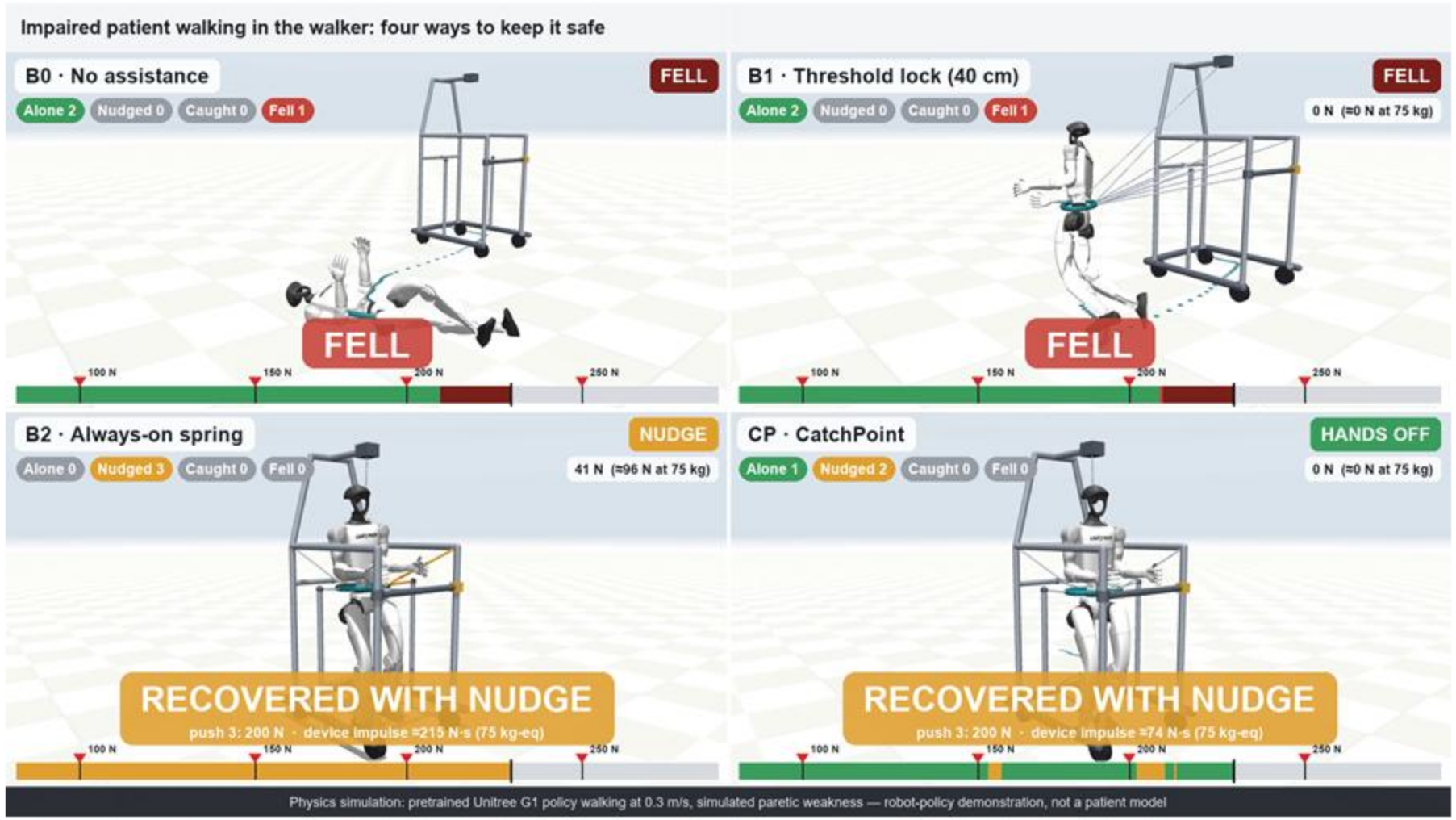
Walking at 0.3 m/s, no pushes.

Patient: right-leg torque ×0.5, 60 ms delay, stiff paretic knee (stronger than the push-test setting).

Paretic knee bends 36° vs 46° healthy.

Falls on its own in 13/20 walks; healthy 0/20.

Walking in the walker: four ways to keep it safe



Robot-policy demonstration only — not patient evidence. Click the picture to play.

Source: MuJoCo physics demo, Unitree G1 policy (walking, v3 protocol).

Walking patient, same pushes: no robot, threshold lock, always-on spring, CatchPoint.

Held-out test, 48 pushes

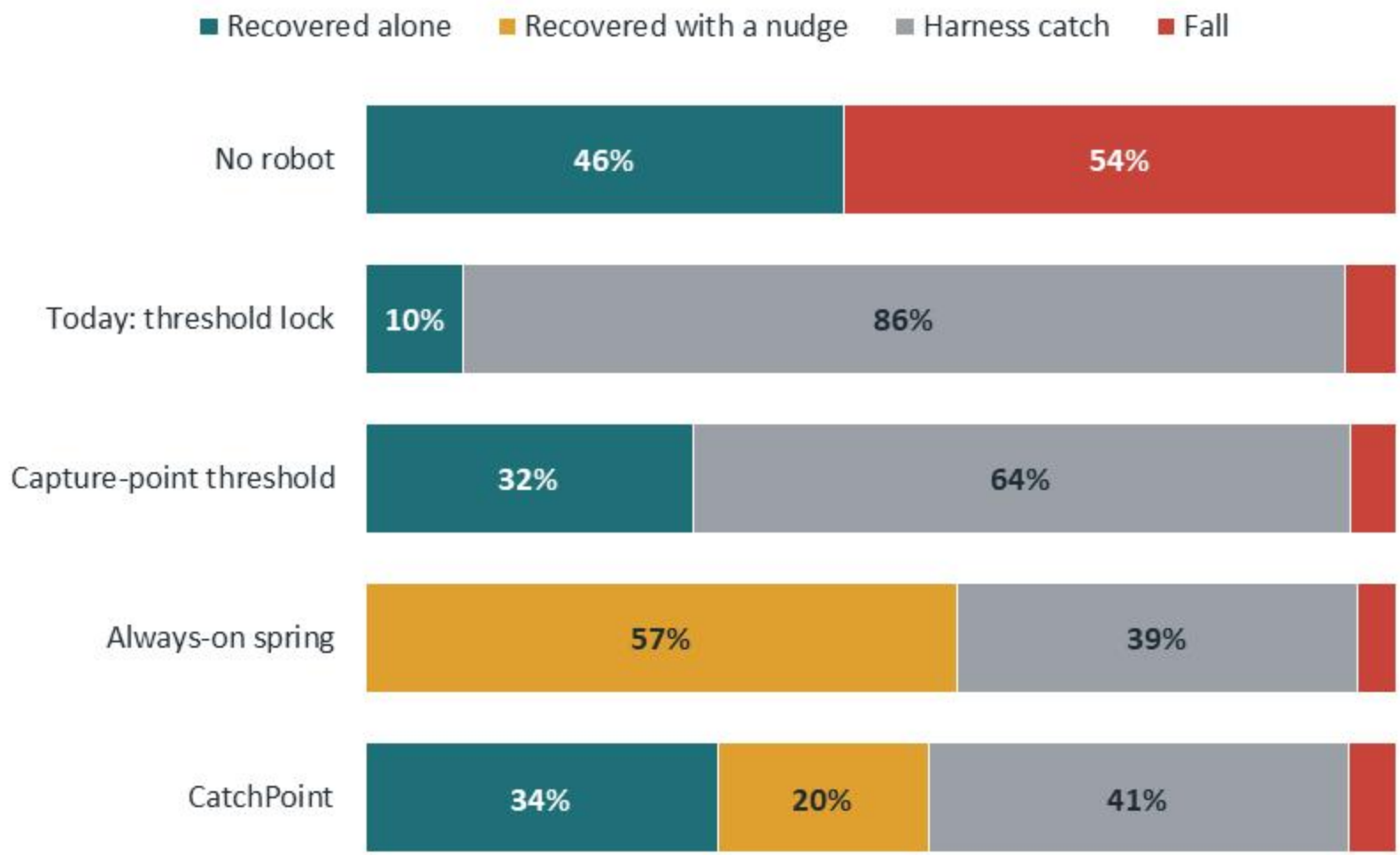
CatchPoint: 0 falls, 7 locks, 33 nudged, 8 alone

Threshold lock: 5 falls, 12 locks, 31 alone

No robot: 17 falls, 31 alone

Trade-off: no falls, but CatchPoint nudges often.

Vs threshold triggers: same falls, fewer and gentler catches



Recovered, no harness catch

54.6% vs 9.6%

34.2% alone + 20.4% nudged, vs 9.6% for threshold lock; both ~5% falls.

vs capture-point threshold

harness -23.0 pp, falls +0.3 pp (n.s.)

Median catch force 175 N vs 382–481 N.
Paired 95% CIs; simplified model.

[Try it: interactive demo ↗]

Source: team Monte Carlo simulation, 2,000 trials per controller, seed 5412. Selected operating points: threshold lock 0.05 m, capture-point threshold 0.08 m, spring 200 N/m, CatchPoint margin 0.03.

Limitations and next steps

What we have not shown

1 Simplified model; CatchPoint is the most model-sensitive: falls 4.8% → 7.0% if steps fall 15% short

2 Physics demo: robot policy; CatchPoint still fell in 9/20 no-push walks (trips)

3 Hardware unbuilt: statics only; base drive, outriggers and spool parts not selected

4 No patient data: all results are simulation

Next steps

1 Bench-test one dual-mode spool

2 Handle forward trips; test other speeds and impairments

3 Clinician feedback on thresholds and stop rules

4 Pilot study (next slide) after ethics approval

Closing the loop on the claims made in this talk.

Pilot evaluation plan: feasibility and safety (proposed)

Proposed protocol — not conducted

Participants

Chronic stroke

Stands ≥ 30 s unaided

Able to step

n = 10–12

n: design target

Randomised within-subject crossover

Threshold lock 2 PBT sessions	→	washout	→	CatchPoint 2 PBT sessions
CatchPoint 2 PBT sessions	→	washout	→	Threshold lock 2 PBT sessions

Order randomised; each person does both

Every session

- Falls (target 0)
- Harness catches
- Self-initiated recovery steps
- Assisted recoveries
- Robot impulse

Before / after each period

- Berg Balance Scale
- Mini-BESTest
- Reactive step latency and length
- FES-I (fear of falling)

Safety

- Stop: any injury
- Stop dose: ≥ 2 unplanned catches in a row
- Therapist spotter
- Ethics approval (IRB)

Targets (n, falls, stop rule) are design proposals

Question

Primary: self-initiated recovery steps per session, CatchPoint vs threshold lock; safety: falls and catches.

Berg / Mini-BESTest are exploratory: two sessions per arm cannot change them.

Learning carries over: order is randomised and period effects are tested.

Ethics approval and a therapist spotter first.

Figure by the project team. Sample size, fall target and stop rule are design proposals. No participants have been recruited.

Summary

1

A new trigger

Act when recovery is no longer feasible for this patient — not when a fixed threshold trips.

2

A graded response

Hands off, minimum nudge, catch only when needed; the same cables are designed to do all three.

3

Evidence so far

In simulation, at similar falls: harness 40.7% vs 63.7% (capture-point threshold) and 85.5% (threshold lock). Hardware: design analysis only.

A Clinical · B Mechanical · C Sensing · D Control · E Simulation

Public page: https://chuanhan1013.github.io/catchpoint/

Interactive simulator demo ↗

Appendix

Technical depth for anyone who wants it

Control law · Simulation model · Results · Robustness · Design D3 analysis · Prior work · Physics-demo protocol · References

A1 · Control law: predict, check, push no more than needed

Where must the foot land? (capture point)

$$\xi = x + \dot{x} / \omega$$

$\omega = \sqrt{(g / z_0)}$; x, $\dot{x}$ = centre-of-mass position and velocity.

Where will it end up with push F?

$$\xi_{land}(F) = p + (\xi - p)\, e^{\omega T} + \beta F$$

ξ = predicted capture point; p = centre of pressure; T = time to foot landing; $\beta = (e^{\wedge}\omega T - 1)/(m\omega^2)$.

Smallest push that keeps the step reachable

$$F^* = \operatorname{argmin} |F|^2 \quad \text{s.t.} \quad h(F) = r - |\xi_{land}(F) - a| \geq 0$$

r = the patient's step reach minus a margin; a = stance foot. Closed-form solution.

Which level?

- h(0) ≥ 0: patient recovers alone, F = 0
- |F*| ≤ 150 N: command F*
- |F*| > 150 N: command 150 N and catch

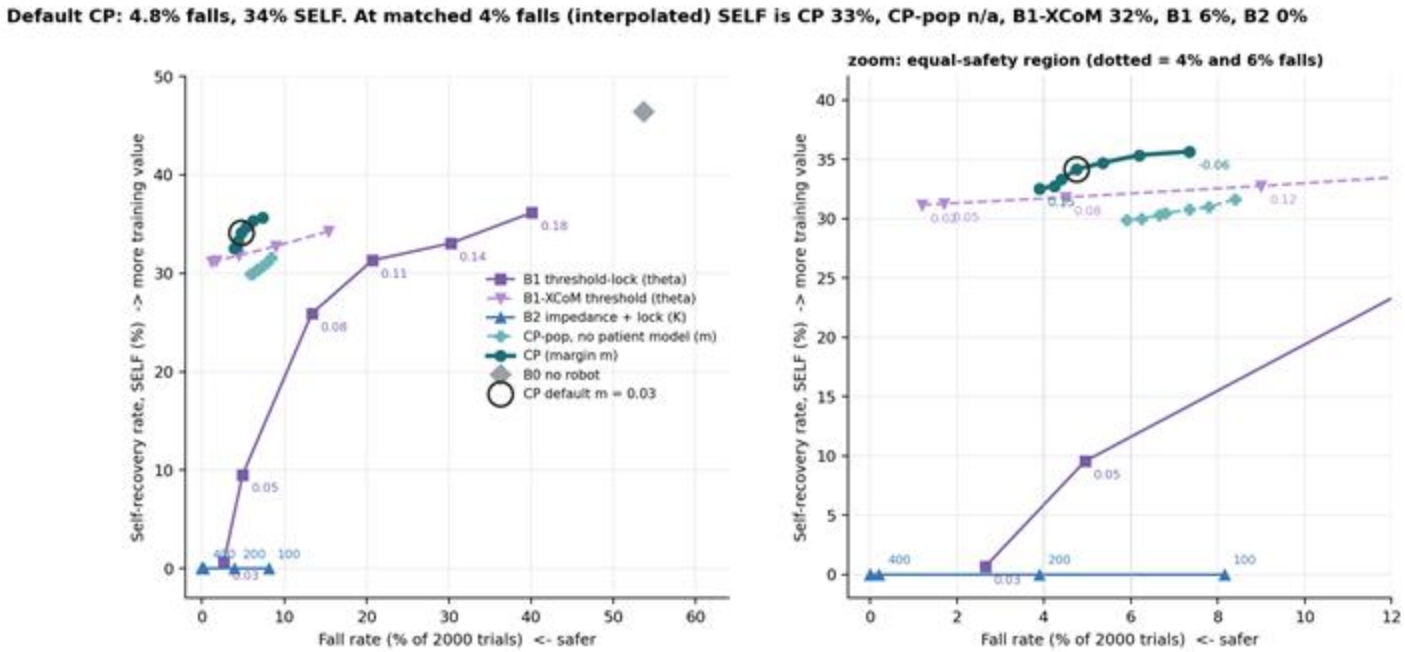
Sources: Hof et al., J Biomech 2005; Pratt et al., Humanoids 2006. Minimum-norm form: a standard safety filter (cf. Ames et al. 2017 [12]); the reachable-step constraint and escalation are ours.

A2 · Simulation model and virtual patients

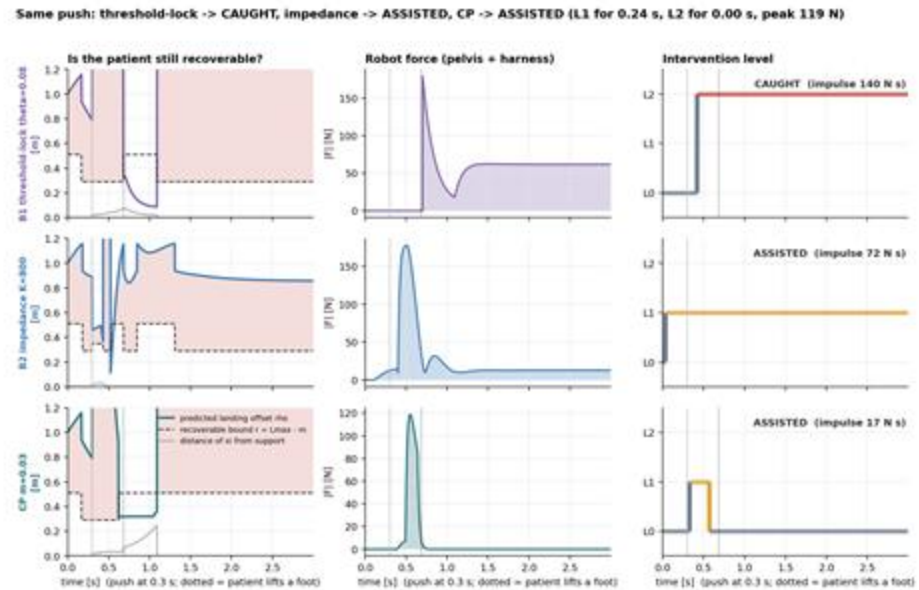
Parameter	Value or range	Note
Balance model	3D linear inverted pendulum	Horizontal plane, 2 ms step
Trial	3.0 s; push at 0.3 s	Δv 0.2–0.9 m/s, any direction
Population	50 patients × 40 pushes	2000 trials; same set for every controller (seed 5412)
Severity s	uniform 0–1	Drives all patient limits below
Mass / COM height	60–85 kg / 0.85–1.00 m	Sampled per patient
Reaction time	0.20 + 0.30·s	Seconds
Max step (paretic / other)	0.50 − 0.30·s / 0.60 − 0.10·s	Metres
Swing time (paretic / other)	0.40 + 0.25·s / 0.35 + 0.10·s	Seconds
Robot delays	sensing 30 ms; actuation 70 ms + 20 ms lag	Catch lock delay 0.28 s
Limits	Assist ≤ 150 N; fall = COM 0.25 m beyond support	Catch: harness spring-damper

All ranges are modelling assumptions anchored to literature, not clinical data. s = severity.

A3 · More practice kept at the same fall rate



Each controller traces a trade-off between falls and self-recovery as its setting is swept.



One showcase push: CatchPoint waits, nudges, and escalates only when needed.

Source: team Monte Carlo simulation, seed 5412.

A4 · Fair comparison at matched fall rates, with confidence intervals

Solo recoveries, interpolated at equal falls

Controller	Solo recoveries at 4% falls	at 6% falls
Threshold lock (B1)	5.9%	11.6%
Capture-point threshold	31.7%	32.1%
Always-on spring (B2)	0.0%	0.0%
CatchPoint (CP)	32.6%	35.2%
CP, population model	n/a (not bracketed)	29.9%

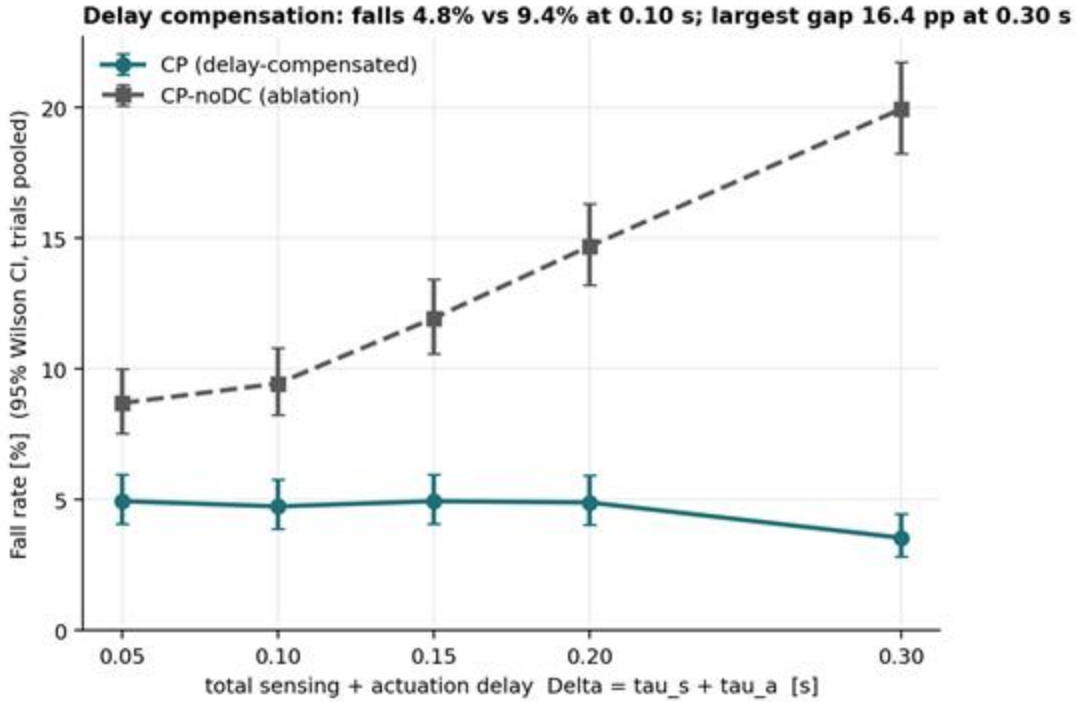
95% patient-bootstrap confidence intervals

Setting	Falls [95% CI]	Solo [95% CI]
Threshold lock 0.05	5.0% [3.4–6.6]	9.6% [7.2–11.9]
Spring 200 N/m	3.9% [2.9–4.9]	0.0% [0.0–0.0]
CatchPoint 0.03	4.8% [3.4–6.2]	34.1% [31.3–37.0]
CatchPoint 0.15	3.9% [2.7–5.3]	32.5% [29.8–35.0]

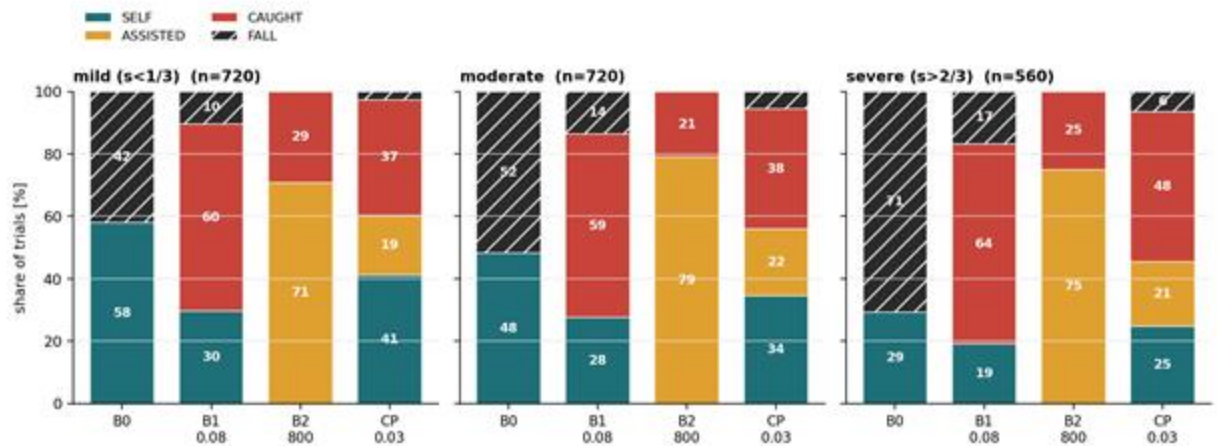
Ablation: the patient-specific model (CP) versus a population-average one (CP-pop, margin 0.03) is worth about 2 pp fewer falls and 4 pp more solo recoveries.

Source: team Monte Carlo simulation, seed 5412. Interpolated between swept settings.

A5 · Robustness: sensing delay and patient severity



Without delay compensation falls rise with latency; with it they stay flat.



Outcomes by patient severity (B0 none, B1 threshold lock 0.08, B2 spring 800, CP = CatchPoint margin 0.03). In the severe group CatchPoint falls 6% and keeps 25% self-recovery.

Source: team Monte Carlo simulation, seed 5412.

A6 · Where the gain comes from, and how fragile it is

CatchPoint minus …	Capture-point threshold	Threshold lock
Falls (pp)	+0.3 [-1.4, 2.0]	-0.2 [-2.3, 2.0]
Harness catches (pp)	-23.0 [-26.7, -19.9]	-44.8 [-49.4, -40.7]
Recovered alone (pp)	+2.4 [1.4, 3.4]	+24.6 [22.3, 27.1]
On own feet (pp)	+22.8 [20.0, 25.8]	+45.0 [41.7, 48.5]

Catch profile	CatchPoint	Capture-pt thr.	Threshold lock
Catch decided after push (median)	0.47 s	0.11 s	0.08 s
Harness force (median)	175 N	481 N	382 N
Falls after a catch decision	83/95	90/90	99/99

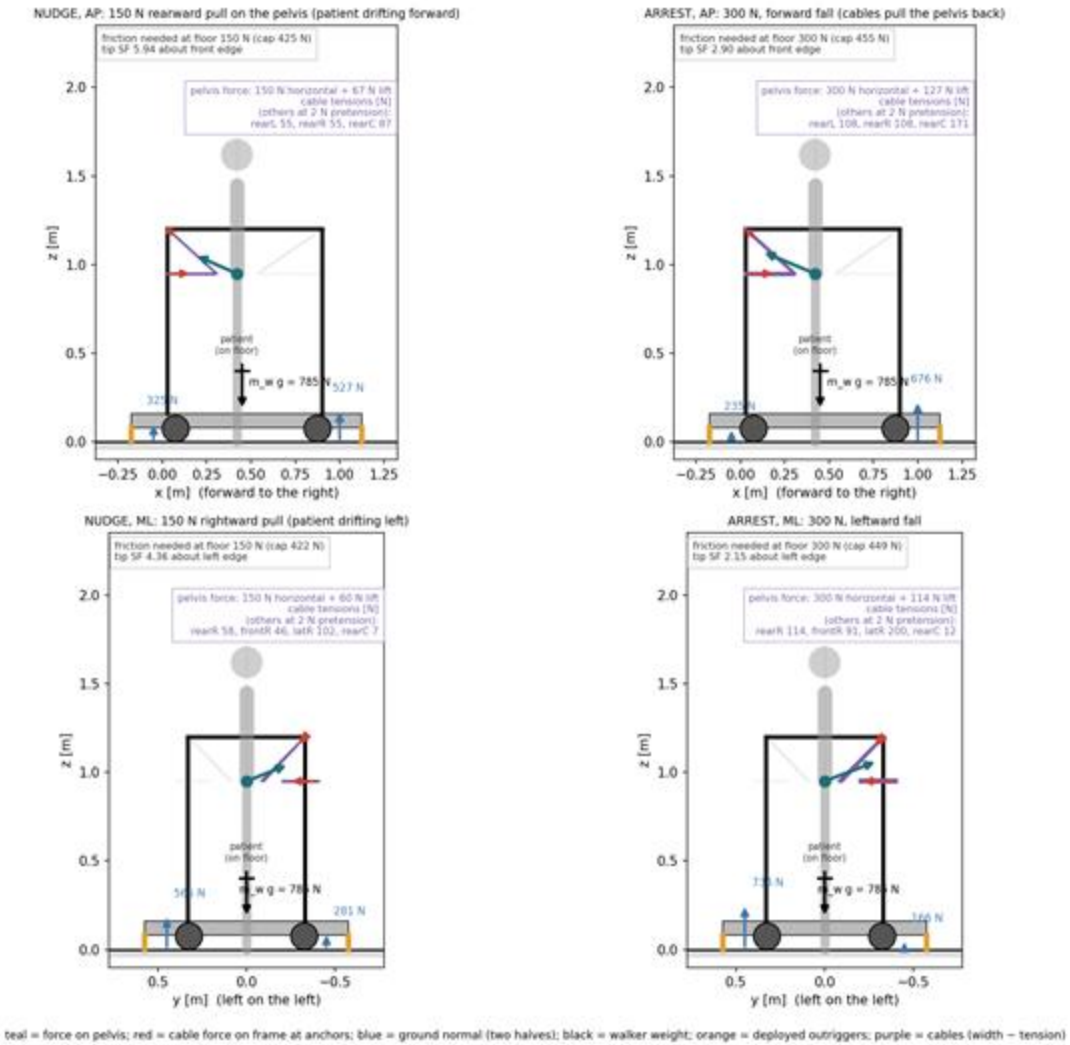
Falls under model error	CatchPoint	Capture-pt thr.	Threshold lock
Nominal (±10% parameters)	4.8%	4.5%	5.0%
Steps 15% short, swing 15% slower	7.0%	4.5%	5.0%
±30% parameter error	12.3%	5.3%	5.0%

Reading. Most of the solo-recovery gain over a threshold lock comes from using the capture point at all; CatchPoint's own edge is the same falls with about a third fewer, gentler catches.

Cost. Catches come late by design; as with the thresholds, most remaining falls (83/95) are locks that came too late. 178 of 408 nudges were not needed. It is the controller most sensitive to model error.

Source: team Monte Carlo, 2,000 pushes, seed 5412; patient-cluster bootstrap (2,000 resamples). [ ] = 95% CI. Sensitivity runs: errors unknown to all controllers.

A7 · Free-body diagrams: nudge and arrest



Nudge, 150 N: tip safety factor 5.94 (AP) and 4.36 (ML); lift on the pelvis ≤ 66.1 N.

Arrest, 300 N, controller-set tensions: tip safety factor 2.90 (AP) and 2.15 (ML); friction needed 300 N against a cap of 455 N (AP).

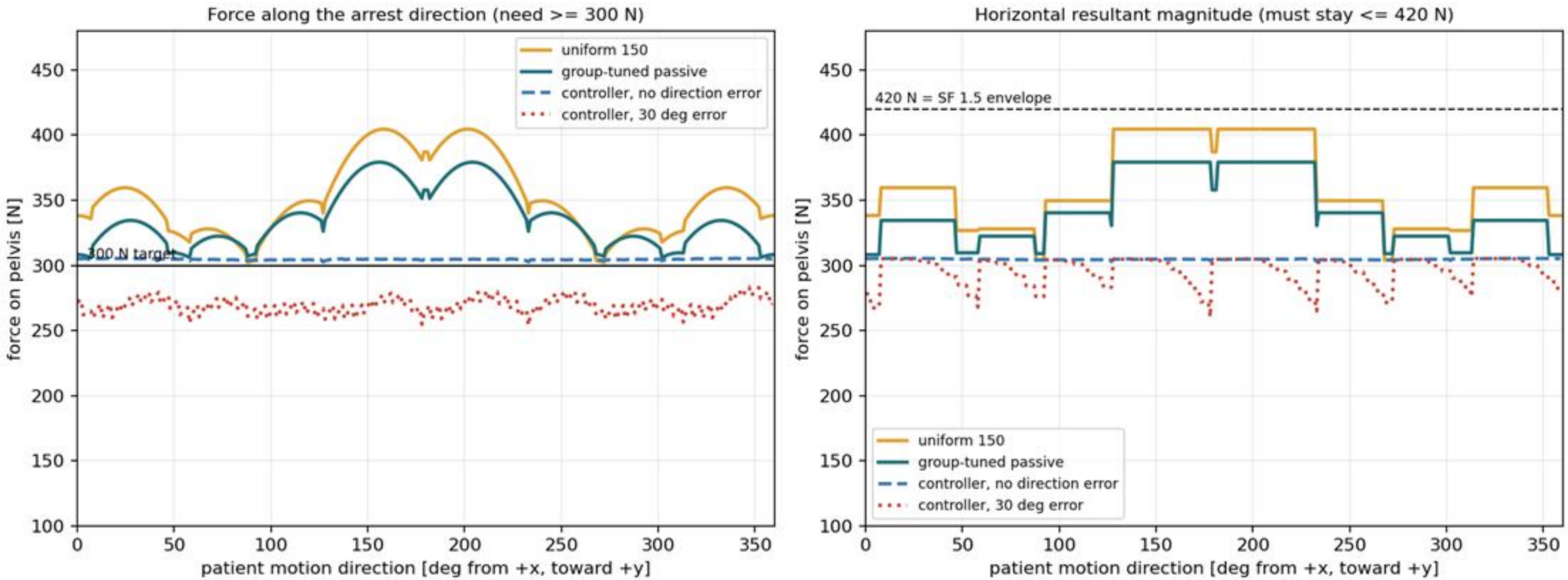
Passive slip brakes (next slide): ≥ 300 N along the fall, ≤ 380 N total; tip SF ≥ 2.09 and sliding SF ≥ 1.28 at the full resultant.

Pelvis lift: 110.4–126.1 N at 300 N (≤ 171 N under passive slip), below the 294 N cap (0.5 body weight of 60 kg).

Statics only; outriggers deployed, mu = 0.5, rigid frame.

Source: team statics analysis of design D3. Unbuilt and unvalidated.

A8 · Passive slip brakes: ≥ 300 N along the fall, ≤ 380 N total



Group-tuned thresholds (corner 150, lateral 155, centre 120 N): ≥ 300 N along the fall (305 N minimum over 360 motion directions, nominal pelvis), worst-case horizontal resultant 380 N, below the 420 N static envelope (SF 1.5; SF 1.0 at 640 N). The perpendicular component reaches 178 N (28° off the fall); independent slipping does not preserve force direction. Tip SF ≥ 2.09, sliding SF ≥ 1.28 at the full resultant; the walker slides at 465 N (259 N if mu = 0.3).

Source: team statics analysis of design D3. Unbuilt and unvalidated.

A9 · Arrest coverage and lock-delay sensitivity

Lock delay	Caught trials	D3, passive slip brakes (headline)	D3, fixed 300 N along the fall (old metric)	Original design, fixed 300 N
0.28 s (baseline)	814	805/814 = 98.8% (CI 97.9–99.5)	804/814 = 98.7% (CI 97.7–99.4)	377/814 = 46.3% (CI 42.9–49.8)
0.15 s	844	821/844 = 97.2% (CI 95.9–98.2)	820/844 = 97.1% (CI 95.8–98.1)	170/844 = 20.1% (CI 17.5–23.0)
0.08 s	866	817/866 = 94.3% (CI 92.5–95.7)	807/866 = 93.1% (CI 91.3–94.7)	56/866 = 6.4% (CI 5.0–8.4)

Headline coverage uses the force the passive slip brakes actually produce for each trial's fall direction: the along-fall component in the stopping model, the full resultant in the static-tip, slide and dynamic-tip checks. Arrested = within 0.25 m, no dynamic tip, static tip SF ≥ 1.5, slide ≤ 1 cm. Wilson 95% CI.

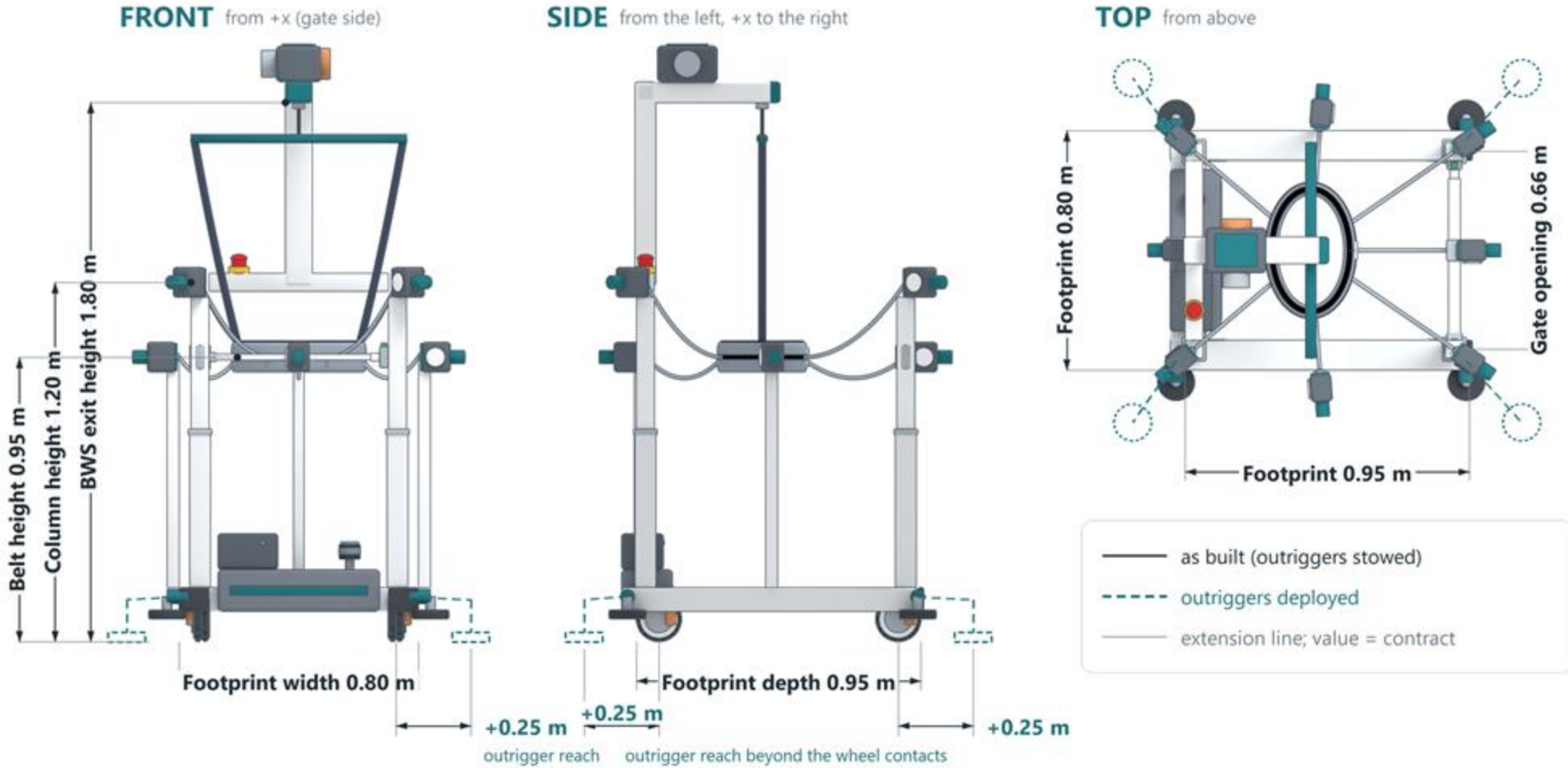
Old metric, lock 0.28 s, fixed along-fall force of 200 / 300 / 400 N: 752 / 804 / 806 of 814. Allowing 491 N is worse: 775/814 = 95.2%, because the static SF-1.5 check fails above 420 N.

Baseline falls 95/2000; the 9 non-arrested caught trials could add to that: 104/2000 at worst. Not a fall-reduction claim.

COM stopped a median 0.104 s after engagement (p95 0.19 s; descriptive, nearest), 814 trials at lock 0.28 s. Slip-brake energy ≤ 22.2 J.

Source: team statics and simulation analysis of design D3. One seeded 2,000-trial run; empirical and conditional on the D3 assumptions.

A10 · Design D3: orthographic views and key dimensions



Belt at nominal position, cables slack; outriggers drawn deployed (dashed). Dimensions come from the shared D3 contract. Concept render: unbuilt and unvalidated.

Render: Blender (design D3 CAD).

A11 · Design D3 specification: geometry, spool, brake and servo

Geometry (from the D3 contract)	Value
Arrest force (horizontal)	passive slip brakes: ≥ 300 N along the fall, ≤ 380 N total (305 N minimum over 360 directions)
Static tip	SF ≥ 1.5 required; envelope 420 N (SF 1.5), 640 N (SF 1.0); SF ≥ 2.09 at the full slip resultant
Walker mass and CG	80 kg incl. ~20 kg low ballast; CG (0.45, 0, 0.4) m
Wheel contacts	x = 0.075 / 0.875, y = ±0.325 m (footprint 0.95 × 0.8 m)
Outrigger feet	+0.25 m all round, deployed on catch; rubber pads, mu ≥ 0.5
Anchors (cable exits)	4 corner at z = 1.20 m; 2 lateral and 2 centre at z = 0.95 m
BWS anchor (cantilever)	(0.42, 0, 1.8) m above the pelvis; hang load 1251 N
Pelvis and belt	pelvis (0.42, 0, 0.95) m; belt half-depth 0.13 m, half-width 0.205 m
Front gate arm	40×20×2 mm 6061-T6 over 0.66 m, latched; cable 150 N, yield SF 4.68; arms only when latched

Spool, brake and servo	Value (basis)
Passive slip thresholds	corner 150 N, lateral 155 N, centre 120 N (spring-applied, power-off brakes)
Slip torque at the spool	3.75 / 3.88 / 3.00 N·m (threshold × spool radius 25 mm)
Rated hold torque	5.63 / 5.82 / 4.50 N·m (1.5 × slip torque)
Worst cable tension, 300 N arrest	199.6 N (controller-set case, worst single cable, 24 directions)
Nudge	150 N resultant, T_max 101.1 N per cable; servo torque 2.53 N·m at the spool; tip SF ≥ 4.36
BWS spool brake	46.91 N·m; hang load 1251 N (1.5 × 85 kg × g, overhead, no horizontal load)
Slip-brake energy	≤ 22.2 J (16.7 J at 0.28 s; arrested trials, three lock delays)
Pelvis lift caps	arrest 294 N (≤ 171 N under passive slip); nudge 88 N (0.5 BW and 0.15 BW of 60 kg)

Not validated: no stress analysis, spool inertia or backlash; mu, cable stiffness and servo data are assumptions. Geometry comes from one contract shared by the CAD and the analysis (generated CAD parts are checked against it to 1 mm).

Source: team statics analysis of design D3 (x forward, y left, z up).

A12 · Closest prior work and how CatchPoint differs

Work	What it does	How CatchPoint differs
RobUST (Omofuma 2023)	Cable pelvis robot; restoring force outside a fixed, subject-specific region	Predictive capture point, step-reach region, escalation to a catch
MRBA / DRBA (NTU)	Pelvic robots; sensor thresholds trigger a brake lock	Preserves the patient's own recovery step; graded force
ExoRecovery (Orhan 2023)	Exoskeleton plans and executes the recovery step	Patient keeps stepping; robot only helps
Monaco APO (2017)	Hip exoskeleton torque once a deviation is detected	Walker interface; recoverable-region logic
mTPAD (Stramel 2023)	Cable pelvic robot on a rollator, about 35 N per cable	Recoverability-based control; conceptual slip-limited catch mode

"To our knowledge": no work found that combines all three contributions; absence of evidence from a search is not proof.

Sources: references [1]–[6]. Some entries were read from abstracts or summaries only.

A13 · Physics-demo protocol, calibration, and limits

Setting	Falls	Catch-lock	Nudged	Hands-off
No robot	32	–	–	24
B1 θ = 0.08 m	0	46	–	10
B1 θ = 0.11 m	0	41	–	15
B1 θ = 0.14 m	0	37	–	19
B1 θ = 0.18 m	1	31	–	24
CP r0 = 0.10 m	0	24	27	5
CP r0 = 0.14 m (default)	0	26	23	7
CP r0 = 0.20 m	0	33	14	9
CP r0 = 0.28 m	0	33	9	14

- 7 push magnitudes (60–300 N) × 8 gait phases; all pushes toward the weak side; forces mass-scaled to a 75 kg equivalent.
- B1 thresholds: θ=.08/.11/.14/.18 m. CP regions: r0=.10/.14/.20/.28 m. Both use the same delays, 0.28 s lock, cable model, and harness catch.
- PPO training used 9.6M steps of unassisted stochastic pushes; the same 56-grid test reduced B0 falls 32→3, with no held-out evaluation.

Kinematic walker; no cable/spool dynamics or collisions; harness torque is assumed; 'caught' denotes simulated catch-lock engagement.

Source: MuJoCo physics demo, Unitree G1 policy (calibrated XCoM region, residual-based escalation).

A14 · Walking demo (v3): held-out test and no-push walks

Pushed while walking	Alone	Nudged	Caught	Fell	Impulse N·s
Healthy, no robot	46	0	0	2	0
Patient, no robot	31	0	0	17	0
Threshold lock	31	0	12	5	20
Always-on spring	0	47	1	0	620
CatchPoint	8	33	7	0	87

No pushes	Walks with a fall	Locks	Walked (m)
Healthy, no robot	0/20	0	8.7
Patient, no robot	13/20	0	9.0
Threshold lock	13/20	0	9.0
Always-on spring	0/20	15	0.8
CatchPoint	9/20	1	9.4

Held out: settings fixed on even push timings × 100–250 N; tested on 48 pushes at odd timings, incl. 125 and 225 N never used for tuning.

Patient: right-leg torque ×0.6, 40 ms delay, walking at 0.3 m/s. Impulse: mean device impulse per push, 75 kg-equivalent.

0/48 falls is compatible with a true rate up to about 6% (95%).

20 walks × 30 s, stronger patient: torque ×0.5, 60 ms delay, stiff knee, motor noise.

Controllers not re-tuned. Falls are mostly forward trips, which CatchPoint was not tuned for; the spring avoids falls but the patient barely walks.

Source: MuJoCo physics demo v3, Unitree G1 policy; one robot, one push direction, deterministic. Robot-policy demonstration, not patient evidence.

A15 · References and image credits

[1] Mobile Robotic Balance Assistant (MRBA), J. NeuroEng. Rehabil., 2023.

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[11] World Health Organization, "Falls," fact sheet.

[12] A. D. Ames et al., "Control barrier function based quadratic programs for safety critical systems," IEEE Trans. Autom. Control, 62(8), 2017.

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